AGENTIC AI · RETAIL INTERVIEW CASE

Returns Desk

Two AI agents handle a store's returns and exchanges on one laptop, with the internet off. The agents make every choice; plain code checks each step and can only say no. Nine questions, answered with measured results.

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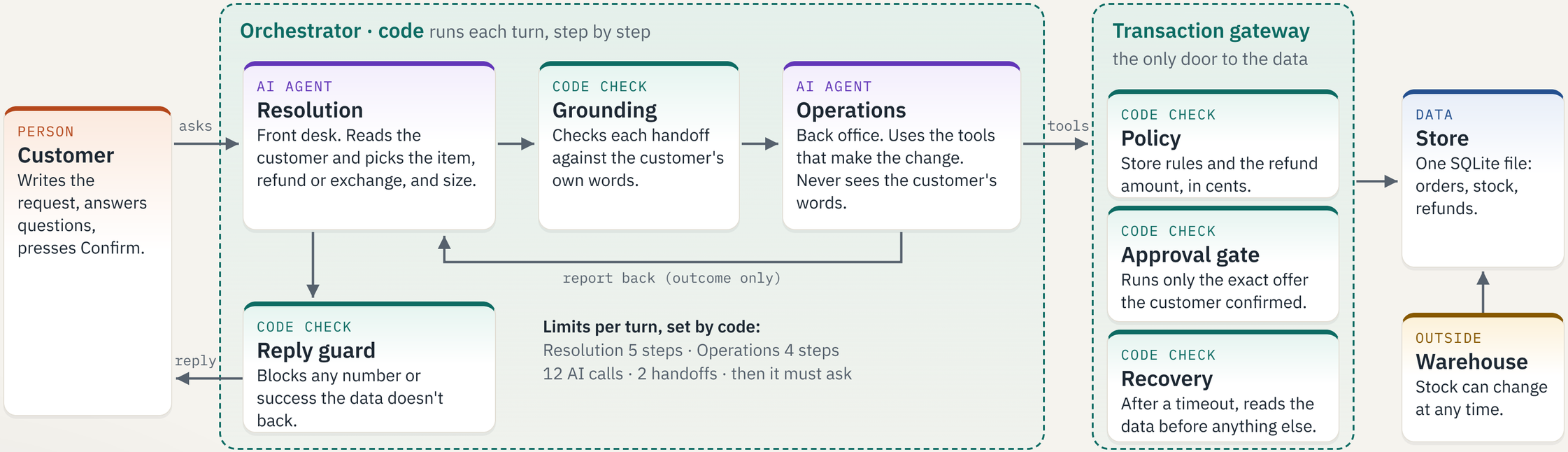
Ali · November 2026

Qwen3.5-4B · llama.cpp · Apple M5 Pro · offline

Q1 · ARCHITECTURE · 2 AI AGENTS, 7 CODE PARTS

Two AI agents do the work; code keeps it safe

✓**Selected: two AI agents with separate jobs. Why:** the agent that reads the customer can't change money or stock, and the agent that can never reads the customer's words. Code can block, read or replay, but never proposes or carries out anything itself.



All on one laptop, internet off: both AI agents run on Qwen3.5-4B (4-bit) with llama.cpp on the Apple M5 Pro graphics chip

AI agent: an AI model that decides the next step and uses tools, not just writes text · **Tool**: one action an agent can ask for, like "create proposal" · **Proposal**: the offer shown to the customer before anything changes · **Store rules**: delivered, unopened, not final sale, within 30 days · **SQLite**: a small database kept in one file on the laptop · **Timeout**: an answer that never arrives · **Block, read, replay**: the only three things code may do to an agent's choice

Q2 · ONE REQUEST, PART 1 · FROM MESSAGE TO OFFER

The agents prepare an offer; nothing changes yet

1 · Orchestrator

Message arrives

Check: a repeat returns the saved result

→

2 · Resolution

Thinks

Reads the chat and account card; picks exchange, line L1, size L

→

3 · Grounding

Checks the handoff

Check: item and size match the customer's words, or retry

→

4 · Operations

Prepares the offer

Never sees the customer's text

Check: rules, stock, amount

→

5 · Resolution

Shows the offer

Check: the reply guard; then Confirm appears

End of turn 1: decision "propose"; money and stock unchanged.

Q3 · ONE REQUEST, PART 2 · FROM CONFIRM TO SAVED RESULT

One button, one exact offer, one save

6 · Customer + approval gate

Confirms

Tied to this exact offer; "yes, but the jeans" runs nothing

→

7 · Operations + gateway

Carries it out

Re-checks, then saves all at once, with a "done" entry

→

8 · Store + Resolution

Reports the truth

The result comes from the database, not the AI

Answer lost after the save (timeout)

1. Saved, but the answer is lost
2. No blind retry: status check only
3. Recovery finds the record: completed
4. Confirm again: same result, no second refund

Q4 · DO THE AGENTS DO THE WORK? · MEASURED

Without the agents, code completes zero refunds

✓ **The agents add 78–83 points of fully correct cases.** Code alone completes 0 refunds or exchanges.

Test set	With AI agents	Code alone	Added
Public (6 cases, 3 runs each)	18/18	1/6	**+83 points**
Variants	241/241	54/241	**+78 points**
Red-team	56/57	11/57	**+79 points**
Blind	58/60	9/60	**+82 points**

Only the agents

Read messy messages, pick item and size, ask, call tools in order, recover.

Code alone

Passes only "ask" or "no" cases. 0 rule breaks in any run.

Q5 · MODEL AND ENGINE

A small open model, on one laptop, internet off

> ✓ **Selected: Qwen3.5-4B in 4-bit, run by llama.cpp on an Apple M5 Pro laptop. Why:** about 5 seconds per case, reliable at filling the agents' tool forms, and no customer data ever leaves the machine.

Choice	Selected	In plain words	Why, and the tradeoff	Evidence
AI model	**Qwen3.5-4B**	An open AI model with 4 billion learned numbers	Fast and reliable at tool calls here; a bigger model may read odd wording better, but slower	Not compared
Model size	**4-bit (B: 8-bit)** `Q4_K_M · B: Q8_0`	Each learned number stored in fewer bits	22% faster and 1.6 GiB less graphics memory, same quality	✓ Measured
Tool calls	**A fixed answer form (B: the model's own format)**	The model can only write answers in the exact shape our tools expect	36% faster and fewer broken answers, same quality	✓ Measured
Engine	**llama.cpp**	Free software that runs the model on the laptop's graphics chip	Works offline, enforces the answer form, reports its own speed; other engines not tested	Not compared
Hardware	**Apple M5 Pro, 24 gigabytes**	One laptop: 15 CPU cores, 16-core GPU, shared memory	No hardware purchase; plugged in, one request at a time	Disclosed

The main tradeoff: speed and privacy over a bigger model's language range. Typos and Spanish product words make the desk ask instead of act (Q9).

Exact run (adapter results.json): unsloth/Qwen3.5-4B-GGUF · Qwen3.5-4B-Q4_K_M.gguf · revision e87f176479d0855a907a41277aca2f8ee7a09523 · sha256 00fe7986ff5f6b463e62455821146049db6f9313603938a70800d1fb69ef11a4 (sha256 source: fresh on first read, then cache(path,size,mtime)) · llama.cpp llama-server 0.7.0 (build 11669, commit 620d2225c) · engine check: warn (strict in the benchmark)

Open model: an AI model anyone can download and run · **Qwen3.5-4B**: the open model used, 4 billion learned numbers · **4-bit / 8-bit** (Q4_K_M / Q8_0): how precisely each number is stored; fewer bits = smaller and faster · **B**: the baseline version · **Engine** (llama.cpp): the program that runs the AI model · **Fixed answer form** (JSON schema): the shape every answer must fit · **GPU**: the graphics chip, which does AI maths in parallel · **CPU**: the general-purpose processor · **GiB**: 2^30 bytes

Q6 · WHEN THINGS GO WRONG · TESTED

Every failure in the brief ends safely

✓ **All 6 public scenarios pass in 3 of 3 runs.** No refund is claimed without a saved record.

Failure	What the desk does	Proof
Unclear request	Asks which item; nothing changes	Public 04
Policy says no	Code blocks it; the reply says why	Public 03
Stock changes	Re-checked at Confirm; offer cancelled	Public 05
Timeout after save	Status check finds the saved refund	Public 06
Repeated approval	Same saved result; no second refund	Public 06
Changed at Confirm	Nothing runs; the new item is offered	Blind set

Q7 · HOW IT WAS TESTED

Every case scored against the right answer

364 test cases

- Public: 6, run 3 times
- Variants: 241 reworded
- Red-team: 57 tricky
- Blind: 60, written without the code

Success means all of

- Every decision right
- Exact offer: item, size, amount
- Final money and stock right
- Zero rule breaks

Fair comparison

- Same laptop, runs interleaved
- Fresh model server each run
- Checker written apart from the desk
- Baseline run twice for noise

What the blind test caught

"Yes, refund the t-shirt" on a jeans offer refunded the jeans. Only comparing final data caught it. Fixed in two rounds: 56 → 57 → 58 of 60.

Q8 · BASELINE VS FINAL · MEASURED

57% faster, with the same quality

✓ **Selected: the final version (F), five changes, each kept only if quality held. Why:** a typical case takes 5.0 s instead of 11.5 s, with the same 18/18 and 1.6 GiB less graphics memory. The baseline already scored 18/18, so it is a fair starting point, not a weak one.

MEASURE	BASELINE	FINAL	CHANGE
Typical case	11.5 s	5.0 s	−56%
Slowest case	13.2 s	5.7 s	−56%
Writing speed (tokens per second)	52	74	+43%
Graphics memory	4.7 GiB	3.1 GiB	−34%
Fully correct	18/18	18/18	same
Overall, per case			**57% faster**

CHANGE IN F	IN PLAIN WORDS	FASTER BY	VERDICT

Fixed answer form
`JSON-schema tool calls`
`B: native`
	The model fills a form instead of free-writing; fewer words	36%	Kept

4-bit model
`Q4_K_M weights`
`B: Q8_0`
	A smaller copy of the same model	22%	Kept

Code writes the report
`code-built ops report`
`B: free text`
	Code sums up tool results, so the AI writes less	18%	Kept

Account loaded up front
`prefetched account card`
`B: off`
	The agent skips one lookup	10%	Kept

Smaller prompt memory
`--cache-ram 2048`
`B: 8192 (MiB)`
	Caps reused-prompt storage at 2 GiB	none	No effect

Each change was tested by putting it back alone; every row kept 18/18. The effects overlap, so they don't add up to 57%. The last change did nothing measurable, and we say so.

Baseline (B): a working version with the same desk code · **Final** (F): B plus five changes · **Typical**: the median, half of cases are faster · **Slowest**: the worst of 18 runs · **Tokens per second**: tokens (word pieces, about ¾ of a word) written per second · **Overall**: the average speed-up when each case is compared with itself (95% range 54–60%) · **Fixed answer form**: the exact shape the model must answer in · **Prompt memory** (prompt cache): reuse of text the model has already read

Returns Desk · 9 · public suite, 6 cases × 3 runs, fresh server per run, laptop plugged in · docs/RESULTS.md §2–§5

Q9 · LIMITATIONS AND NEXT STEPS

Tested limits fail safe; one known gap is left

Limitation or idea	Today	Status
Typos or Spanish product names	Asks which item instead of acting	Seen in tests
A misspelled change at Confirm the agent also misses	The confirmed offer still runs	Known gap
English only	Other languages get more questions	Known
Every test set used for tuning	Hidden cases are the real test	Known
Typo-tolerant, multilingual product list	Would close the first two rows	Proposed
Blind set written by people	A true unseen test	Proposed
Speculative decoding (a helper model)	Could cut writing time, the biggest cost	Proposed